

2018 Spring		Dept. or School		proctor	
Calculus-I Midterm-Exam		Student ID		Name	
1. (3pts) Evaluate $\lim_{x \rightarrow 1^+} (\ln x^2) \tan(\frac{\pi}{2}x)$.		3. (3pts) Evaluate $\frac{dy}{dx}$ at $(x,y)=(1,1)$, if the two variables x and y satisfy $x^2y^2 + xy - 2 = 0$.			
2. (3pts) Evaluate $f'(\sqrt{2})$ for $f(x) = \arctan(\frac{1+x}{1-x})$.		4. (3pts) Evaluate $\int_1^{e^5} t^4 \ln t \, dt$.			

5. (3pts) Let A be the amount of paint needed for coating a hemisphere dome with diameter 100m in the thickness of 0.05cm. Estimate the value $\frac{A}{\pi}$ by using the differential.

6. (4pts) Evaluate

$$\int_0^2 \frac{x^3}{(x^2+4)^{3/2}} dx .$$

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7. (4pts) Evaluate $\int_{\frac{\sqrt{2}}{3}}^{\frac{2}{3}} \frac{1}{x^3 \sqrt{9x^2 - 1}} dx$		8. (5pts) Let R be the region bounded by the curve $y = \sinh x$, the x-axis and the line $x = \ln 10$. Find the volume of the solid which is obtained by rotating R around the line $x = -2$.			

9. (6pts) Evaluate

$$\int_0^1 \frac{t+1}{t^2+t+1} dt$$

10. (6pts) Starting at a point x_0 on a line, a particle moves along the line with velocity

$$v(t) = \sin^2 t \cos 3t$$

at time t . Find the maximum distance from x_0 to the position of the particle.